

EC2

- 1-What is an EC2 instance type, and how do you choose the right one for your application?
- 2-What is an EC2 instance family, and when would you use one family over another?
- 3-Describe the typical steps involved in launching an EC2 instance.
- 4-What is an EC2 user data script, and how can it be used during instance launch?
- 5-Explain the purpose of EC2 instance metadata and how you can access it from within an instance.
- 6-How can you create custom AMIs, and why might you want to do so?
- 7-What are security groups, and how do they control inbound and outbound traffic to EC2 instances?
- 8-Explain the use of Network Access Control Lists (NACLs) and how they differ from security groups.
- 9-How do you enable and configure AWS Web Application Firewall (WAF) in front of an EC2-based web application?
- 10-What is Auto Scaling, and how can it be used to ensure high availability and scalability of EC2 instances?
- 11-Explain the purpose of Amazon Elastic Load Balancing (ELB) and its integration with EC2 instances.
- 12-What is Amazon EC2 Container Service (ECS), and how does it help with containerized applications on EC2 instances?
- 13-How can you configure Amazon Route 53 for DNS-based load balancing of EC2 instances?
- 14-What is status check in EC2 instance?
- 15-How to change instance types for running application without downtime of application?
- 16-What is the difference between AMI and Snapshot?
- 17-How to boot related issues like kernel panic in EC2 instances?
- 18-How many maximum number of IPs can be attached to an instance?
- 19-Describe different types of purchasing options available in AWS?
- 20-What are the different types of AWS Placement Groups, and how do they differ?
- 21-Can you change the placement group of a running EC2 instance?
- 22-What is the difference between an Availability Zone and a Placement Group?

- 23-What are some best practices for using Placement Groups in AWS?
- 24-Explain the limitations or constraints of AWS Placement Groups?
- 25-Give examples of scenarios or applications where each type of EBS volume (gp2, io1, st1, sc1, gp3) is the most appropriate choice.
- 26-What is Amazon Elastic Block Store (EBS), and how does it differ from Amazon S3?
- 27-What are the different types of EBS volumes available, and when would you use each type (e.g., gp2, io1, st1, sc1, gp3)?
- 28-Explain the concept of Provisioned IOPS (PIOPS) and when it's necessary for achieving consistent performance.
- 29-How do you resize an EBS volume, and what precautions should be taken when doing so?
- 30-What is the difference between EBS volume types and EBS volume size, and how do they impact performance?
- 31-What is an EBS snapshot, and why is it important for data durability and disaster recovery?
- 32-How often should you create EBS snapshots, and what strategies can be employed for efficient backup and retention policies?
- 33-What are the best practices for encrypting EBS volumes at rest, and how do you implement encryption?
- 34-Describe the difference between EBS-backed and instance-store-backed EC2 instances and their respective advantages and limitations.
- 35-How can you monitor the performance and health of EBS volumes, and what AWS services or tools can assist in this process?